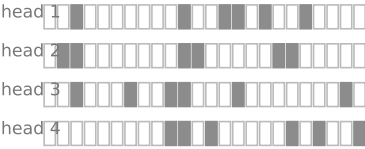


One shared cell array, three eviction classes: only a sequence-level, attention-scored keep-set both compacts and keeps the needle

(a) per-head evictors

SnapKV · Ada-KV · H2O · TOVA



↓ union: a cell frees only if EVERY head dropped it



live = 16/24 cells — holes everywhere

✗ cannot compact — SnapKV retains 67.7% (GPU; 91-100% at 32 heads) · FA-off decode 0.1-0.2x

(b) position-blind, seq-level

StreamingLLM (its own budget)



middle deleted unseen
compacts (contiguous by design)



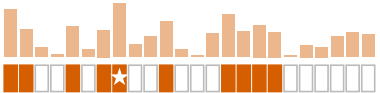
dense — but the needle is gone

△ same speed as μ KV (1.20x), but blind: needle at $\leq 67\%$ depth lost — NIAH 7/14

(c) attention-scored, seq-level

μ KV — one keep-set for all heads & layers

prefill attention (kq_evict side node, $\sim +1\%$ prefill):



in-place slide
(peak memory never rises)



K dense cells — needle kept

✓ compacts to K=1024 $\rightarrow$ 723 live cells (7.4%)
NIAH 14/14 · decode 1.20x GPU / 4.80x CPU